

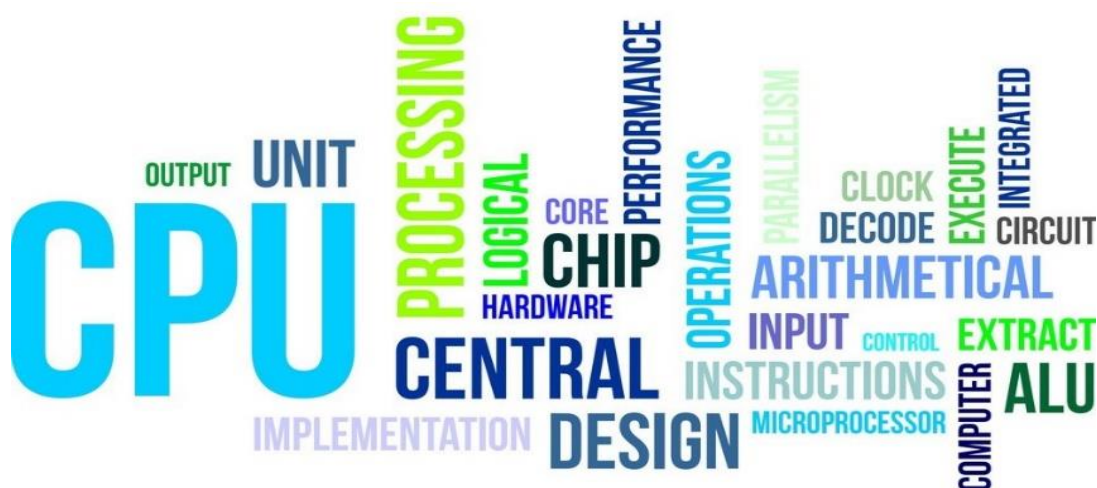
GOVERNMENT POLYTECHNIC PORBANDAR



Affiliated to
Gujarat Technological University, Ahmedabad

Laboratory Manual

MICROPROCESSOR AND ASSEMBLY LANGUAGE
PROGRAMMING (3330705)
D. E. Second Year (Semester– III)



Government Polytechnic - Porbandar

Opp. Airport, National Highway
Porbandar – 360578

GOVERNMENT POLYTECHNIC PORBANDAR



Enrollment No: _____

Name: _____

Address: _____

Subject: _____

Branch: _____

GOVERNMENT POLYTECHNIC PORBANDAR



Computer Engineering Department

Certificate

This is to certify that

Mr./Ms _____ Enrolment No: _____
Of 3rd Semester Diploma course in Computer Engineering has
satisfactorily completed his/her term work in Microprocessor and
Assembly Language Programming (3330705) from _____
to _____ in Government Polytechnic, Porbandar.

Date of Submission: _____

Staff in charge

Head of Department

GOVERNMENT POLYTECHNIC PORBANDAR



VISION OF THE INSTITUTE

To emerge as a leading polytechnic having value based effective technical education system in the region to meet the future demand of industry and society.

MISSION OF THE INSTITUTE

M1: To impart adequate technical skill and capability to accept challenges and to adopt rapidly changing scenario in technological development as well as to take up higher studies.

M2: Foster close networking with the industry, alumni and the guardians.

M3: Promote professional ethics and human values, integrity, accountability and transparency.

M4: Promote entrepreneurship practices to generate employment opportunities.

M5: Create conducive academic environment.

GOVERNMENT POLYTECHNIC PORBANDAR



COMPUTER ENGINEERING DEPARTMENT

VISION OF THE DEPARTMENT

To achieve excellence in Computer Engineering by imparting technical and problem-solving skills along with ethical value to meet industrial requirements having social and environmental concerns.

MISSION OF THE DEPARTMENT

M1: To provides a learning ambiance to enhance discipline knowledge, technical skill and problem solving skill.

M2: To motivates students for lifelong learning to adapt challenges in rapidly changing technology.

M3: To induces ethical values and spirit of social commitment.

M4: To provide opportunities to promote leadership skill required in Computer Engineering industry's diverse culture.

Subject: Microprocessor and Assembly Language Programming

Subject Code: C205, 3330705

Course Outcomes:

CO1: Explain architecture of Intel 8085 microprocessors.

CO2: Describe the bus structure, timing diagram and memory Organization of 8085 Microprocessors.

CO3: Develop Assembly Language programs using 8085 Instructions.

CO4: Describe the basics of interrupts and its applications.

CO5: Explain architecture of Intel 8086 microprocessors.

List of Practicals

Sr. No.	Practical Title	Mapping with COs	Mapping with POs, PSOs
1	1) To study 8085 microprocessor. 2) To study Pin diagram of 8085 microprocessor. 3) Demonstrate of kit/simulator of 8085.	CO1	PO1
2	1) Implement program to perform addition of two 8-bit numbers using 8085. 2) Implement program to perform subtraction of two 8-bit numbers using 8085.	CO3	PO1, PO2, PO3, PO4
3	1) Implement program to perform mask the lower four bits of content of the memory location. 2) Implement program to perform set higher four bits of content of the memory location to 1.	CO3	PO1, PO2, PO3, PO4
4	1) Implement program to perform Exclusive OR of two numbers. 2) Implement program to perform exchange the content of two memory locations.	CO3	PO1, PO2, PO3, PO4
5	1) Implement program to perform add 16-bit numbers. 2) Implement program to perform subtract 16-bit numbers.	CO3	PO1, PO2, PO3, PO4
6	Implement program to perform the multiplication of two 8-bit numbers using 8085.	CO3	PO1, PO2, PO3, PO4
7	Implement program to perform the division of two 8-bit numbers using 8085.	CO3	PO1, PO2, PO3, PO4
8	Implement program to check whether given no is odd or even.	CO3	PO1, PO2, PO3, PO4
9	Implement program to sum integers from 0 to 9.	CO3	PO1, PO2, PO3, PO4
10	Implement program to copy content of one memory location to another memory location.	CO3	PO1, PO2, PO3, PO4
11	Implement program to count no of positive, negative and zero value in given block of data.	CO3	PO1, PO2, PO3, PO4
12	1) Implement program to find the smallest number from an array of n number. 2) Implement program to find the largest number from an array of n number.	CO3	PO1, PO2, PO3, PO4
13	1) Implement program to add an array of data. 2) Implement program to calculate the length of a string given.	CO3	PO1, PO2, PO3, PO4
14	Implement program to reverse a string using stack operations and store it in the same memory area.	CO3	PO1, PO2, PO3, PO4

Lab Evaluation Rubrics:

Lab Performance (LP) Rubrics

Criteria	Excellent (4)	Good (3)	Substandard (2)	Poor (1)
Lab Performance (LP)	* Student Performs experiment as per the instructions given without any help from faculty. * Student demonstrates an accurate understanding of the lab objectives and concepts. * Student can explain concepts to fellow classmates.	* Student needs some help from faculty to perform the experiments as per the instruction * Student has sufficient understanding of lab objectives and concepts. * Student can understand concepts by himself but cannot explain it to fellow classmates.	* Student performs of the experiments with the help from faculty. * Student has no clear understanding of lab objectives and concepts. * Student cannot explain concepts to fellow classmates.	* Student cannot complete experiment. * Student has not understanding of lab objectives and concepts. * Student cannot explain concepts to fellow classmates.

Lab Assignment (LA) Rubrics

Criteria	Excellent (4)	Good (3)	Substandard (2)	Poor (1)
Lab Assignment	* Student demonstrates an accurate understanding of the lab objectives and concepts. * Student answers questions completely and correctly. * Programs are written neatly, has no error.	* Student has a basic knowledge of content, but may lack some understanding of some concepts. * Student answers questions fairly well. * Programs could have be written more neatly. * Programs has no errors or errors are minimal	* Student has problems with both the graphs and the answers. * Student appears to have not fully grasped the lab content and the Programs possess some errors.	* Student turns in lab report late or the report is so incomplete and/or so inaccurate. * Programs are not complete.

Index

Sr. No	Practical	Page No	Date	Staff Signature	PA	
					LP	LA
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Practical - 1

AIM:

- 1) To study 8085 Microprocessor.**
- 2) To study Pin diagram microprocessor 8085.**
- 3) Demonstrate of kit/simulator of 8085.**

Introduction

The functional block diagram or architecture of 8085 Microprocessor, gives the complete details about a Microprocessor. It includes the ALU (Arithmetic and logic unit), timing and control unit, instruction registers and decoder, register array, interrupt control, and serial I/O control.

The 8085 microprocessor was made by Intel in mid 1970s. It was binary compatible with 8080 microprocessor but required less supporting hardware thus leading to less expensive microprocessor systems. It is a general purpose microprocessor capable of addressing 64k of memory. The device has 40 pins, require a +5V power supply and can operate with 3MHz single phase clock.

8085 Architecture

The 8085 has a 16 bit address bus which enables it to address 64 KB of memory, a data bus 8 bit wide and control buses that carry essential signals for various operations. It also has a built in register array which are usually labelled A (Accumulator), B, C, D, E, H, and L. Further special-purpose registers are the 16-bit Program Counter (PC), Stack Pointer (SP), and 8-bit flag register F. The microprocessor has three maskable interrupts (RST 7.5, RST 6.5 and RST 5.5), one Non-Maskable interrupt (TRAP), and one externally serviced interrupt (INTR). The RST n.5 interrupts refer to actual pins on the processor a feature which permitted simple systems to avoid the cost of a separate interrupt controller chip.

1. Control Unit

Generates signals within microprocessor to carry out the instruction, which has been decoded. In reality causes certain connections between blocks of the processor be opened or closed, so that data goes where it is required, and so that ALU operations occur.

2. Arithmetic Logic Unit

The ALU performs the actual numerical and logic operation such as 'add', 'Subtract', 'AND', 'OR', etc. Uses data from memory and from Accumulator to perform arithmetic and always stores the result of operation in the Accumulator.

3. Registers

The 8085 microprocessor includes six registers, one accumulator, and one flag register, as shown in Fig. In addition, it has two 16-bit registers: the stack pointer and the program counter. The 8085 has six general-purpose registers to store 8-bit data; these are identified as B, C, D, E, H, and L as shown in Fig. They can be combined as register pairs - BC, DE, and HL - to perform some 16-bit operations. The programmer

can use these registers to store or copy data into the registers by using data copy instructions.

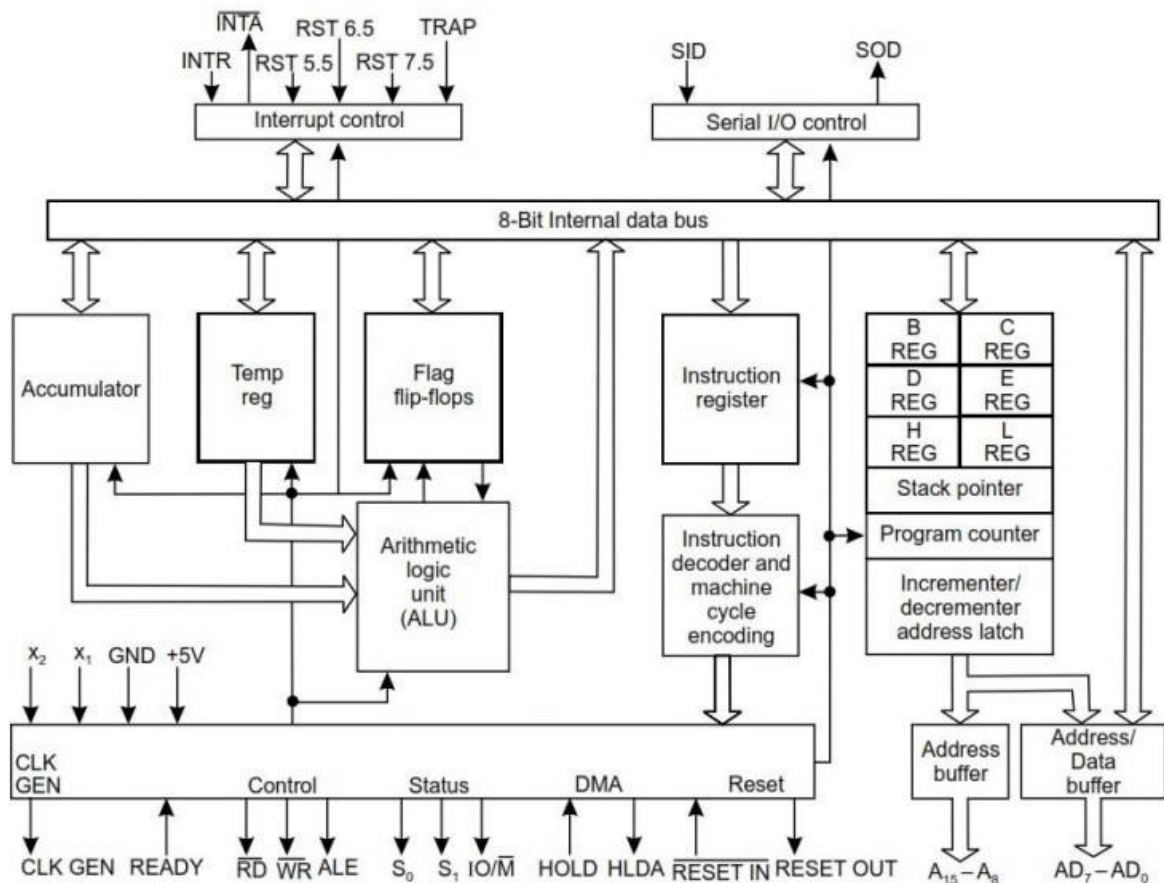


Figure 1 Architecture of 8085

4. Accumulator

The accumulator is an 8-bit register that is a part of arithmetic/logic unit (ALU). This register is used to store 8-bit data and to perform arithmetic and logical operations. The result of an operation is stored in the accumulator. The accumulator is also identified as register A.

5. Flag Registers

The ALU includes five flip-flops, which are set or reset after an operation according to data conditions of the result in the accumulator and other registers. They are called Zero (Z), Carry (CY), Sign (S), Parity (P), and Auxiliary Carry (AC) flags. The most commonly used flags are Zero, Carry, and Sign. The microprocessor uses these flags to test data conditions.

6. Program Counter (PC)

This 16-bit register deals with sequencing the execution of instructions. This register is a memory pointer. Memory locations have 16-bit addresses, and that is why this is a 16-bit register. The microprocessor uses this register to sequence the execution of the instructions. The function of the program counter is to point to the memory address from which the next byte is to be fetched. When a byte (machine code) is being fetched, the program counter is incremented by one to point to the next memory location.

7. Stack Pointer (SP)

The stack pointer is also a 16-bit register used as a memory pointer. It points to a memory location in R/W memory, called the stack. The beginning of the stack is defined by loading 16-bit address in the stack pointer.

8. Instruction Register / Decoder

This is a temporary store for the current instruction of a program. Latest instruction is sent to here from memory prior to execution. Decoder then takes instruction and 'Decodes' or interprets the instruction. Decoded instruction is then passed to next stage.

9. Memory Address Register (MAR)

Holds addresses received from PC for eg: of next program instruction. MAR feeds the address bus with address of the location of the program under execution.

10. Control Generator

Generates signals within microprocessor to carry out the instruction, which has been decoded. In reality it causes certain connections between blocks of the processor to be opened or closed, so that data goes where it is required, and so that ALU operations occur.

11. Register Selector

This block controls the use of the register stack. Just a logic circuit which switches Between different registers in the set will receive instructions from Control Unit.

12. 8085 System Bus

The microprocessor performs four operations primarily.

- Memory Read
- Memory Write
- I/O Read
- I/O Write

All these operations are part of the communication processes between microprocessor and peripheral devices. The 8085 performs these operations using three sets of communication lines called buses - the address bus, the data bus and the control bus.

a. Address Bus

The address bus is a group of 16 lines. The address bus is unidirectional: bits flow only in one direction – from the 8085 to the peripheral devices. The microprocessor uses the address bus to perform the first function: identifying a peripheral or memory location. Each peripheral or memory location is identified by a 16 bit address. The 8085 with its 16 lines is capable of addressing 64 K memory locations.

b. Data Bus

The data bus is a group of eight lines used for dataflow. They are bidirectional: data flows in both direction between the 8085 and memory and peripheral devices. The 8 lines enable the microprocessor to manipulate 8-bit data ranging from 00 to FF.

c. Control Bus

The control bus consists of various single lines that carry synchronization signals. These are not groups of lines like address or data bus but individual lines that provide a pulse to indicate an operation. The 8085 generates specific control signals for each operation it performs. These signals are used to identify a device type which the processor intends to communicate.

8085 Pin Diagram

8085 is a general purpose microprocessor having 40 pin which shown in below figure.

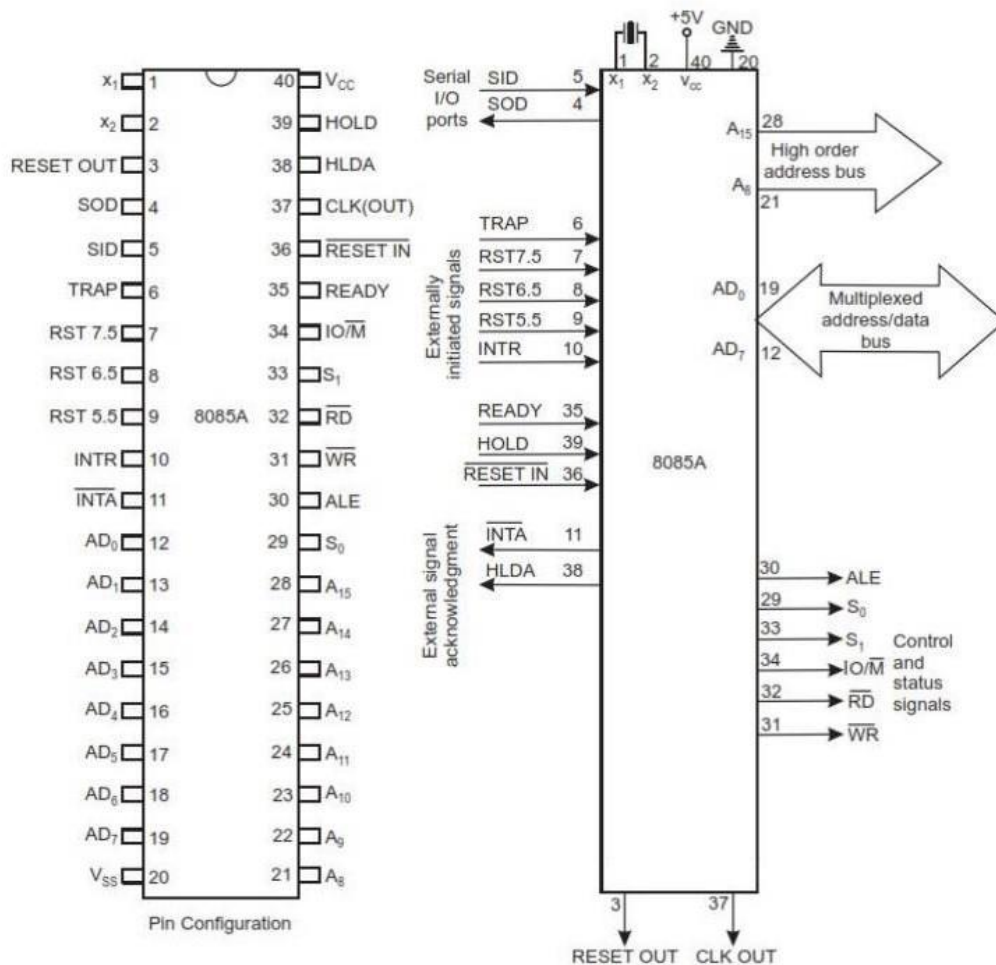


Figure 2 Pin Diagram of 8085

8085 Pin Description

To study the pin diagram we group the signals into 5 categories:

- Power supply and clock signals
- Address bus
- Data bus
- Control and status signals
- Interrupts and externally initiated signals
- Serial I/O ports

1. Power supply and Clock frequency signals:**Vcc:**

+ 5 volt power supply

Vss:

Ground

X1 X2:

Crystal or R/C network or LC network connections to set the frequency of internal clock generator. The frequency is internally divided by two. Since the basic operating timing frequency is 3 MHz, a 6 MHz crystal is connected externally. CLK (output)-Clock Output is used as the system clock for peripheral and devices interfaced with the microprocessor.

2. Address Bus:**A8 - A15 (Output)**

It carries the most significant 8 bits of the memory address or the 8 bits of the I/O address;

3. Data bus:**AD0 - AD7 (input/output)**

- These multiplexed set of lines used to carry the lower order 8 bit address as well as data bus.
- During the opcode fetch operation, in the first clock cycle, the lines deliver the lower order address A0 - A7.
- In the subsequent IO / memory, read / write clock cycle the lines are used as data bus.
- The CPU may read or write out data through these lines.

4. Control and Status signals:**ALE (output) - Address Latch Enable.**

- It is an output signal used to give information of AD0-AD7 contents.
- It is a positive going pulse generated when a new operation is started by uP.
- When pulse goes high it indicates that AD0-AD7 are address.
- When it is low it indicates that the contents are data.

RD - Read memory or IO device

This indicates that the selected memory location or I/O device is to be read and that the data bus is ready for accepting data from the memory or I/O device

WR - Write memory or IO device

This indicates that the data on the data bus is to be written into the selected memory location or I/O device.

IO/M - Select memory or an IO device

This status signal indicates that the read / write operation relates to whether the memory or I/O device.

It goes high to indicate an I/O operation.

It goes low for memory operations.

5. Status Signals:

SO, S1 (Output)

It is used to know the type of current operation of the microprocessor.

Data Bus Status: Encoded status of the bus cycle:

S1	S0	OPERATION
0	0	HALT
0	1	WRITE
1	0	READ
1	1	FETCH

6. Interrupts and externally initiated operations

They are the signals initiated by an external device to request the microprocessor to do a particular task or work.

There are five hardware interrupts called,

- *TRAP*
- *RST 7.5*
- *RST 6.5*
- *RST 5.5*
- *INTA*

On receipt of an interrupt, the microprocessor acknowledges the interrupt by the active low *INTA* (Interrupt Acknowledge) signal.

Reset In (input, active low)

- This signal is used to reset the microprocessor.
- The program counter inside the microprocessor is set to zero.
- The buses are tri-stated.

Reset out (Output)

- It indicates CPU is being reset.
- Used to reset all the connected devices when the microprocessor is reset.

7. Direct Memory Access (DMA): Tri state devices

- When 2 or more devices are connected to a common bus, to prevent the devices from interfering with each other, the tristate gates are used to disconnect all devices except the one that is communicating at a given instant.
- The CPU controls the data transfer operation between memory and I/O device. Direct Memory Access operation is used for large volume data transfer between memory and an I/O device directly.
- The CPU is disabled by tri-stating its buses and the transfer are affected directly by external control circuits.
- **HOLD** signal is generated by the DMA controller circuit.
- On receipt of this signal, the microprocessor acknowledges the request by sending out **HLDA** signal and leaves out the control of the buses. After the HLDA signal the DMA controller starts the direct transfer of data.

READY (input)

Memory and I/O devices will have slower response compared to microprocessors. Before completing the present job such a slow peripheral may not be able to handle further data or control signal from CPU. The processor sets the READY signal after completing the present job to access the data.

The microprocessor enters into WAIT state while the READY pin is disabled.

8. Single Bit Serial I/O ports

These signals are used for serial communication.

SID (Input)

Serial input data line: The data on this line is loaded into accumulator bit 7 whenever a RIM instruction is executed.

SOD (output)

Serial output data line: The output SOD is set or reset as specified by the SIM Instruction.

8085-Simulator

8085 Simulator is powerful application that supplies 8085 microprocessor users and educators with user-friendly graphical development environment for Windows with integrated simulator.

From the 4 bit microprocessor brought out by Intel in 1971, advancement in technology have been made and now 8 bit ,16 bit , 32 bit and 64 bit microprocessors are available and 64 bit and 32 bit microprocessors are dominating the market.

The trainer kit 8085 based training tool developed specifically for learning the operation of today's microprocessor based systems.

8085 simulator supports all 8085 instructions. User enters 8085 assembly language programs, whose simulation is performed by the software.

The purpose of a microprocessor trainer kit is to:

- Facilitate people in learning and using machine code programming.
- Familiarise people of the basic microprocessor hardware.
- Facilitate testing of hand coded programs.
- Facilitate easy interface to external I/O devices.
- Facilitate training with its peripherals also.

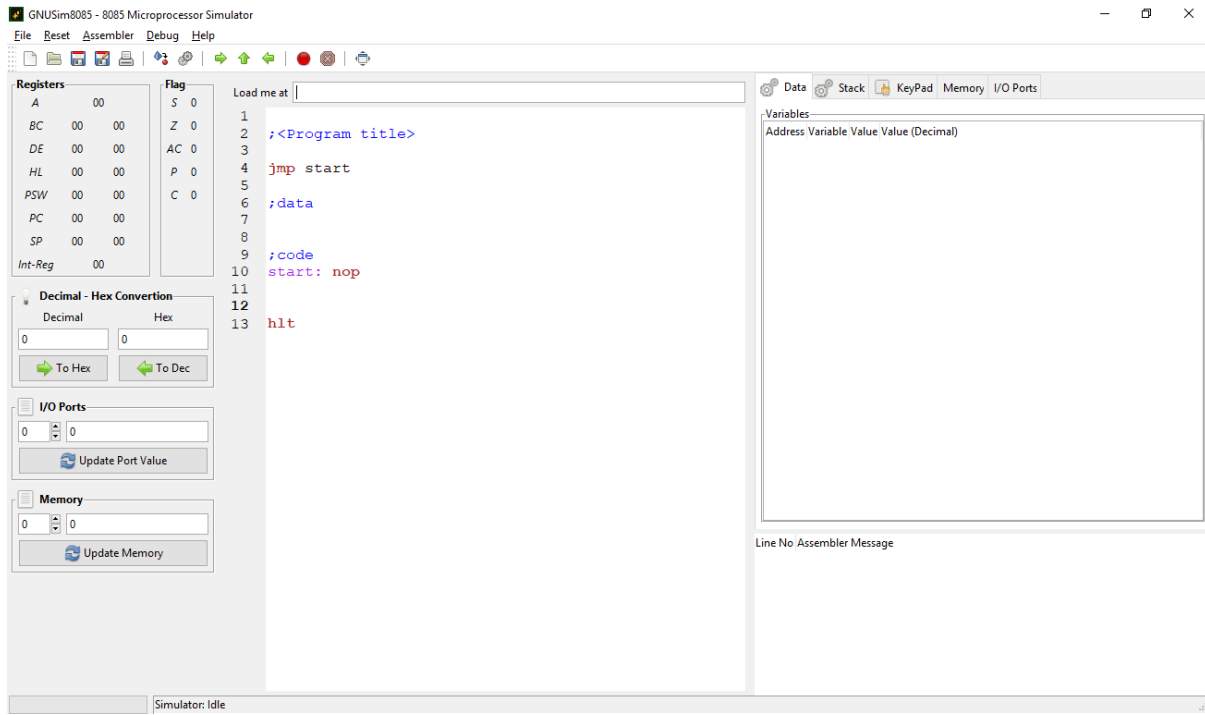


Figure 3 GNU simulator for 8085

Features

(a) Program entry

- Program editor with interactive input wizard for all the standard instructions.
- Set address and verify the contents of present location.
- Data is set altered in memory immediately for each hex key entry.
- A separate opcode view, which displays assembled code in hex.

(b) Assembler Editor

- Can load Programs.
- Supports assembler directives.
- Number parameters can be given in binary, decimal and hexadecimal format.
- Supports writing of comments.
- Supports labelling of instructions, even in macros.
- Has error checking facility.

(c) Memory Editor

- Can directly update data in a specified memory location
- It has 3 types of interface, user can choose from it according to his need.
 - Show entire memory content
 - Show only loaded memory location
 - Store directly to specified memory location
- Allows user to choose memory range.

(d) I/O Editor

- It is necessary for peripheral interfacing.
- Enables direct editing of content

(e) Program execution/Simulator

There are 2 modes of simulator engine:

Run all at a Time: It takes the current settings from the simulation speed level and starts execution accordingly.

Step by Step: It is manual mode of control of FORWARD and BACKWARD traversal of instruction set

Questions

1. What is a Microprocessor?
2. What is the memory size of 8085 microprocessor?
3. 8085 is _____ bit microprocessors.
4. What are the types of buses?
5. What is the function of accumulator?
6. What are the different types of flags in 8085 microprocessor?
7. What are the types of general purpose registers in 8085?
8. What is the size of Data bus?
9. What are the various interrupts in 8085 microprocessor?
10. Which register pairs can be used as 16 bits data registers?
11. Explain use of Program counter.
12. What is Stack Pointer?
13. Stack pointer is a _____ bit register.
14. Program counter is a _____ bit register.
15. Accumulator is a _____ bit register.
16. What is the clock frequency of 8085 microprocessor?
17. Why Address-Data lines (AD0-AD7) are multiplexed?
18. What are the control and status signals of 8085?
19. Explain the functions of ALE and IO/M signals of the 8085 microprocessor
20. What is the function of READY Pin?
21. What is Maskable interrupts?
22. What is Non-Maskable interrupts?
23. Which interrupt has the highest priority?
24. What is the function of TRAP?

Practical - 2

AIM:

- 1) Implement program to perform addition of two 8-bit numbers using 8085.***
- 2) Implement program to perform subtraction of two 8-bit numbers using 8085.***

Practical - 3

AIM:

- 1) Implement program to perform mask the lower four bits of content of the memory location.***
- 2) Implement program to perform set higher four bits of content of the memory location to 1.***

Practical - 4

AIM:

- 1) Implement program to perform Exclusive OR of two numbers.***
- 2) Implement program to perform exchange the content of two memory locations.***

Practical - 5

AIM:

- 1) Implement program to perform add 16-bit numbers.***
- 2) Implement program to perform subtract 16-bit numbers.***

Practical - 6

AIM:

Implement program to perform the multiplication of two 8-bit numbers using 8085.

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Practical - 7

AIM:

Implement program to perform the division of two 8-bit numbers using 8085.

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Practical - 8

AIM:

Implement program to check whether given no is odd or even.

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Practical - 9

AIM:

Implement program to sum integers from 0 to 9.

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Practical - 10

AIM:

Implement program to copy content of one memory location to another memory location.

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Practical - 11

AIM:

Implement program to count no of positive, negative and zero value in given block of data.

Practical - 12

AIM:

- 1) Implement program to find the smallest number from an array of n number.***
- 2) Implement program to find the largest number from an array of n number.***

Practical - 13

AIM:

- 1) Implement program to add an array of data.***
- 2) Implement program to calculate the length of a string given.***

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Practical - 14

AIM:

Implement program to reverse a string using stack operations and store it in the same memory area.

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